02 DATA TYPES AND VARIABLES • DETAILED SOLUTIONS

output solutions

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

Detailed Solutions — Predict the Output & Debugging (Module 02)

- **Q3 Solution**: Unsigned integers obey well-defined modulo 2^N arithmetic (`0U - 1U` wraps to `UINT_MAX`), whereas signed integer overflow is Undefined Behavior.
- **Q4 Solution**: `int sum;` is uninitialized (contains indeterminate value), so `sum += i;` has Undefined Behavior! Initialize `int sum = 0;` .